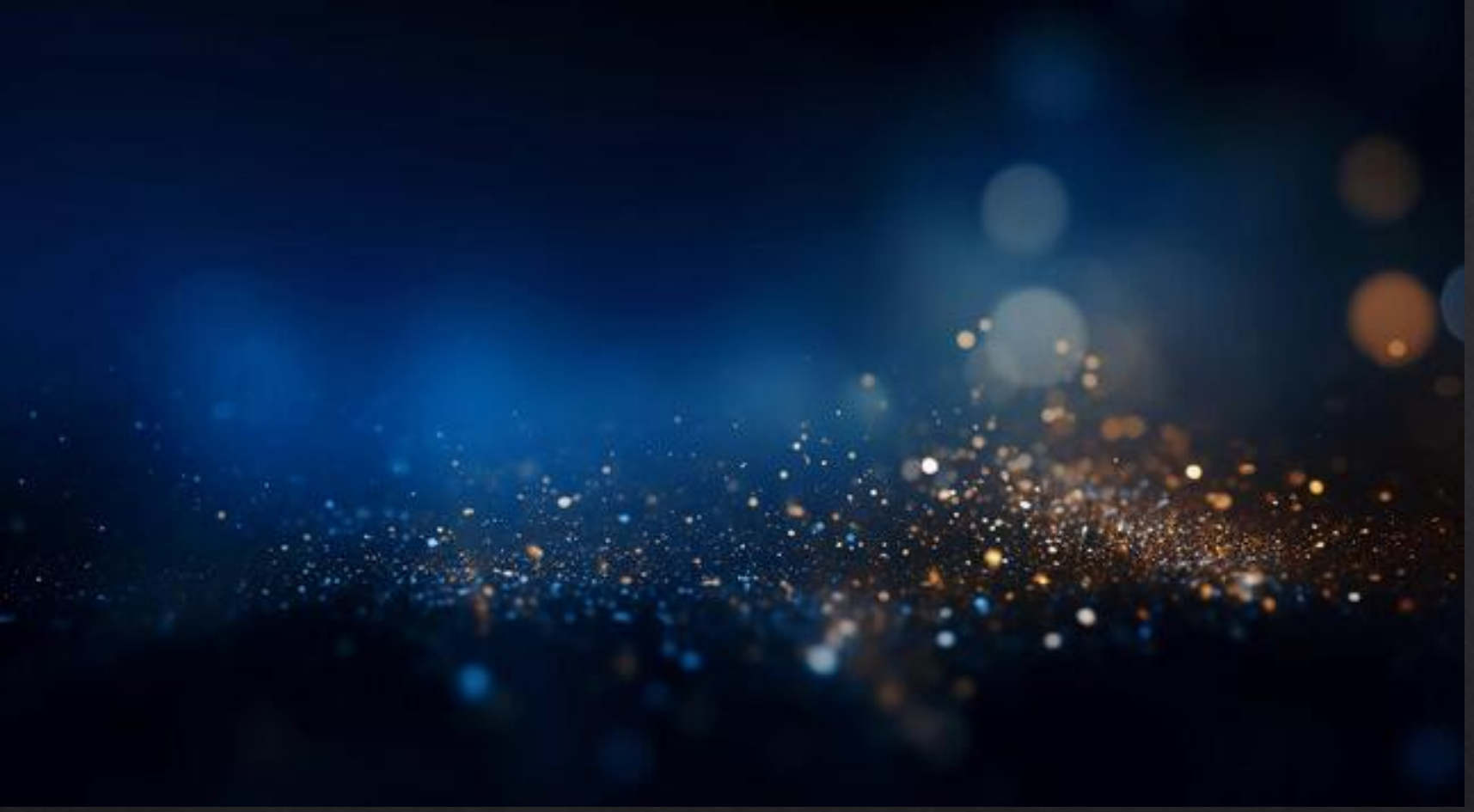


MA416

Lecture 15 – October 23, 2025

Outlier Detection

Dr. Matthew N. Moore

Group Project Details

Step 1: Selected group members

- Group must contain at least 2 people
- Group must contain at most 3 people

Step 2: Obtain dataset(s) that allows the group to conduct hypothesis tests on the structures mentioned in Step 3.

- The dataset(s) must be legitimate, not artificially generated
- Possible locations to download are Kaggle, Usa Census, Us.gov, etc.
- Dataset(s) must contain at least 300 observations and contain at least 2 categorical variables and at least 2 numerical variables

Step 3: General submission requirements

- The final submission must be a PDF file completely generated from/in a R-Markdown format
- The final due date is 11:59pm Sunday November 9, 2025; to be submitted on Canvas

Group Project Details

Step 4: Specific submission requirements (sections)

- Must conduct a one-factor ANOVA test with number of groups at least 4
- Must conduct a repeat-measures ANOVA test with number of groups at least 3
- Must conduct a non-parametric test on a claim of the groups choosing

Step 5: Specific submission requirements (within each section)

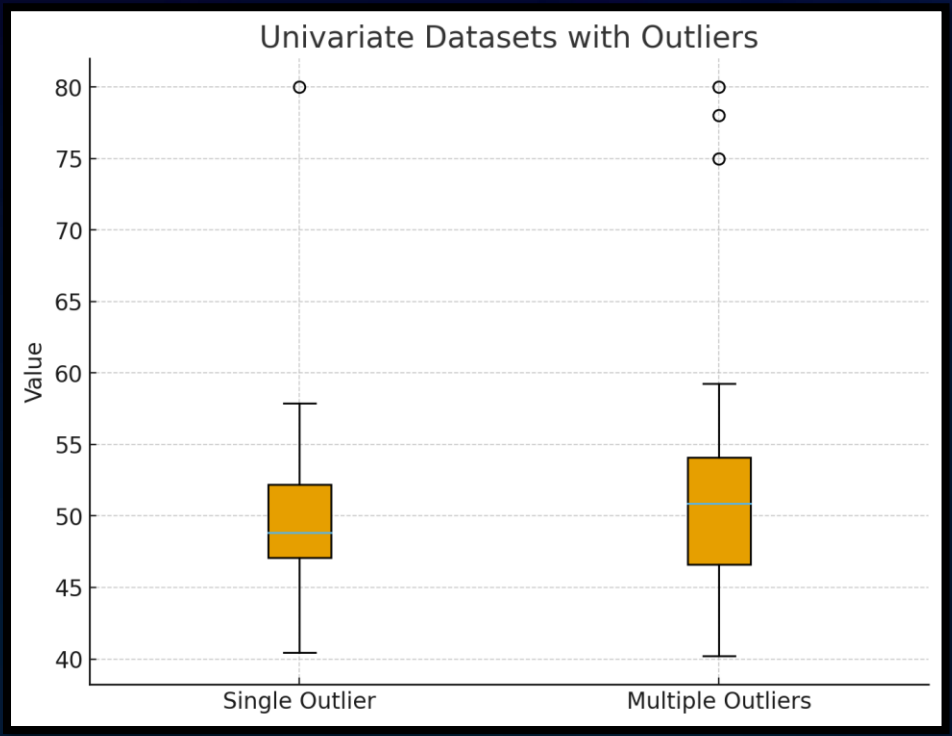
- If a test has any assumptions, the quantitative assumptions must all be statistically tested in a valid manner. All qualitative assumptions (e.g., independence within groups) must at the least be discussed in terms of its (potential) validity
- All null hypotheses must clearly be stated prior to conducting the test, in addition to a discussion on the potential consequences or applications for which the conclusion of such test has regarding the study
- There must be at least one appropriate use of (a) bootstrapping, (b) Monte-Carlo simulation, (c) confidence intervals
- There must be at least three appropriate uses of graphical representations (using base R graphing techniques) including (but not limited to) histograms and/or box plots
- There must be at least six appropriate and meaningful uses of LaTex code for equation representation, not including in-line equations
- All datasets used should be publicly accessible from online (such as a website or GitHub)

Step 6: Grading

- There will be 3 stars associated to this assignment, each of which will be awarded on a hierarchy-point basis. For example, if there are 10 possible points in the assignment, star 1 awarded at 3, star 2 awarded at 7, star 3 awarded at 10.
- There will be at most two submissions. The first submission is optional and must be made in person (not digital) to the professor at the latest by 5pm Tuesday, November 4, 2025; it should be printed and stapled.

What is an Outlier?

Definition: An "outlier" is an observation in a dataset that lies an abnormal distance from other values in the sample, relative to a specified model that approximates or resembles the population for which the sample is extracted from.



It's possible that there is either one or multiple outliers in a data set. There are some things to keep in mind:

1. Outliers depend on model assumptions
2. Removing outliers can distort the dataset and statistics
3. Graphical diagnostics should come first
4. If conducting a hypothesis test or constructing a model, both versions (with and without outliers) should be reported.
5. Always check for a theoretical or practical cause for the existence of the outlier(s).
6. Never remove outliers sequentially.

The Grubbs Test

Suppose you have a dataset S with $|S| = n$ and $x^* \in S$ where $S \subseteq P$. We seek to test:

$$H_0: x^* \text{ is not an outlier for } P$$
$$H_a: x^* \text{ is an outlier for } P$$

We define x^* to be the point such that $|x_k - \bar{x}|$ is maximal when $x_k = x^*$. The Grubbs' test statistic is defined as:

$$G_{stat} = \frac{\max_k |x_k - \bar{x}|}{s_x}$$

One can show (we won't) that G_{stat} follows a modified T distribution with $\nu = n - 2$ degrees of freedom. Let $T = t_{crit}$ be the associated two-tailed critical value associated to a T_{n-2} distribution with $\frac{\alpha}{2n}$ associated to each of the tails. Then:

$$G_{crit} := \left(\frac{n-1}{\sqrt{n}}\right) \cdot \sqrt{\frac{T^2}{T^2 + (n-2)}}$$

We reject the claim provided that $G_{stat} > G_{crit} \gg 0$.

The Grubbs Test

Example: Outlier Detection on Hours Worked

```
mydata = read.csv('CompanyPayroll_9.25.csv')
X = mydata$Hours_Worked

max(abs(X - mean(X))) # 27.3461
```

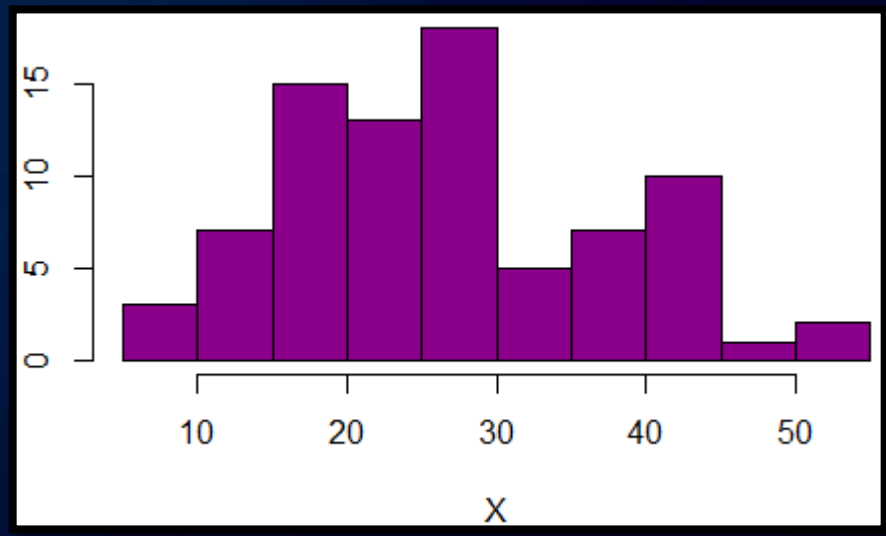
```
xstar = X[which.max(abs(X - mean(X)))]
personstar = mydata$First_Name[which.max(abs(X - mean(X)))]

personstar # Kasim
xstar      # 53.92
```

```
Gstat = ??? # 2.5669

alpha = 0.05
n = length(X)
T = qt(alpha/(2*n), n-2, lower.tail=F)

Gcrit = ??? # 3.3106
```



Since $G_{stat} < G_{crit}$, we do not have evidence to believe that Kasim is an outlier. Moreover, since this individual is the most extreme, we also have evidence to believe that this dataset does not have any outliers.

The Tietjen-Moore Test

Suppose you have a dataset $S \subseteq P$ with $|S| = n$ and define $S^* \subset S$ with $|S^*| = n_o$ be the conjectured outliers of S, typically with $n_0 \ll n$. Then we seek to test that:

H_0: S^* are not outliers for P

H_a: All of S^* are outliers for P

Define S_{or} as $S - S^*$ (the set of values for which the conjectured outliers are removed), where $|S_{or}| = n - n_o$. Then we define the following metrics:

$$\bar{x} = \frac{1}{n} \cdot \sum_{k=1}^{n} (x_k) \quad x_k \in S \qquad\qquad \bar{x}_{or} = \frac{1}{n - n_o} \cdot \sum_{k=1}^{n-n_o} (x_k) \quad x_k \in S_{or}$$

Notice that if S^* are truly outliers, then the variance of S_{or} should be substantially less than that of S. Hence, we define:

$$TM_{stat} := \frac{\sum_{k=1}^{n-n_0} (x_k - \bar{x}_{or})^2}{\sum_{k=1}^{n} (x_k - \bar{x})^2}$$

Theorem: $TM_{stat} \in [0, 1]$.

Question: Is the rejection region associated to TM_{stat} left-tailed, right-tailed, or both?

The Tietjen-Moore Test

Example: Outlier Detection on Average Temperature

```
mydata = read.csv('EnvStudyData_9.25.csv')

X = sort(mydata$AverageTemperature_C)
n = length(X)      # 327
```

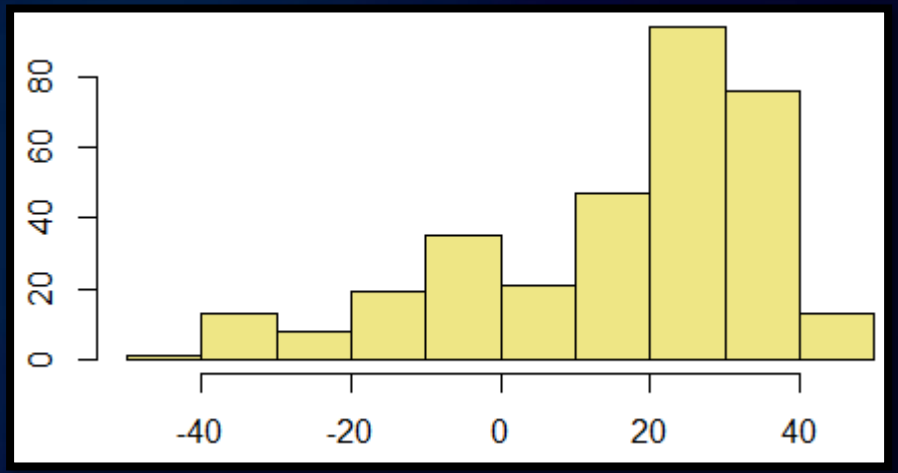
```
Xstar = X[1:5]

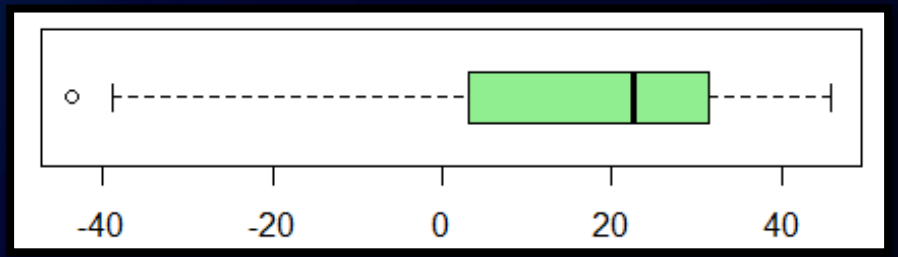
Xor = setdiff(X, Xstar)

xbar = mean(X)
xbaror = mean(Xor)
```

```
TMstat_top = ???      # 117599.2
TMstat_bot = ???      # 135910.2

TMstat = TMstat_top/TMstat_bot # 0.8653
```





Question: We have a test statistic, but what do we compare it to?

The Modified Tietjen-Moore Test

```
TMstat_top = ???      # 117599.2
TMstat_bot = ???      # 135910.2

TMstat = TMstat_top/TMstat_bot # 0.8653
```

Note that since TM_{stat} is bounded on $(0, 1]$ with a left-tailed rejection region, then:

$$TM_{stat}^{mod} = -\ln(TM_{stat})$$

will be supported on $[0, \infty)$ with a right-tailed rejection region. Ideally, we would like there to be some $c > 0$ such that $c \cdot TM_{stat}^{mod} \sim \chi_{\nu}^2$ for some ν. One can show that:

$$\mathbb{E}\left(TM_{stat}^{mod}\right) \approx \ln\left(\frac{n-1}{n-n_0-1}\right) \qquad \text{and} \qquad \mathbb{V}\left(TM_{stat}^{mod}\right) \approx \frac{2}{n-n_0-1} + \frac{2}{n-1}$$

for large n. Hence, one can define:

$$Z_{stat}^{\delta} := \frac{-\ln(TM_{stat}) - \ln\left(\frac{n-1}{n-n_0-1}\right)}{\sqrt{\frac{2}{n-n_0-1} + \frac{2}{n-1}}}$$

```
Zstat_top = ???
Zstat_bot = ???

Zstat = Zstat_top/Zstat_bot
Zstat # 1.1624

pval = 2*pnorm(Zstat, lower.tail=F)
pval  # 0.2451
```

```
> Xstar
[1] -43.58 -38.87 -38.65 -38.04 -37.54
```

Since $p_{val} > \alpha$, we have evidence to believe that S^* are not outliers of S.